ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C6 · MECHANICAL SYSTEMS

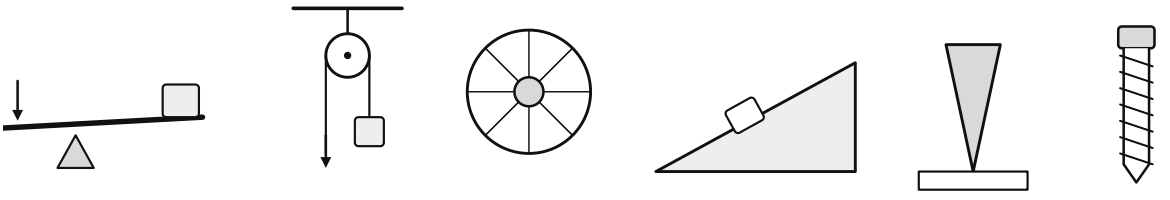
# Six simple machines

★☆☆
Grade 5

**Learning goals**

- Name and identify simple machines.
- Label a lever.

**1** **Write the letter of each simple machine.** [6]

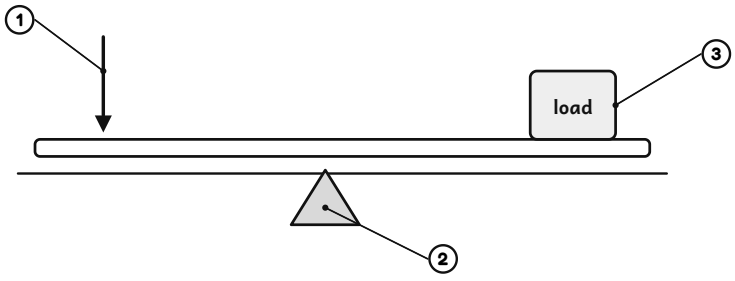



| Machine | Letter |
|---|---|
| wedge | E |
| lever | A |
| screw | F |
| pulley | B |
| inclined plane | D |
| wheel and axle | C |

**2** **Label the lever.** [3]

LABELS

**fulcrum (pivot)** **load** **effort (push)**




① effort (push) ② fulcrum (pivot) ③ load

**3** **Sort the examples into the simple machine they use.** [9]

SORT THESE

**dheki** **axe** **bottle opener** **scissors** **door knob** **screwdriver** **rickshaw wheel** **knife** **boti**

| Lever | Wedge | Wheel and axle |
|---|---|---|
| scissors | axe | rickshaw wheel |
| bottle opener | boti | screwdriver |
| dheki | knife | door knob |

**4** **Fill in the gaps.** [4]

WORD BANK

**gravity** **friction** **direction** **more** **less** **pivot**

A lever turns on a ___pivot___. A pulley can change the ___direction___ of a force. A ramp lets you lift a load with ___less___ force. Brakes stop a bicycle using ___friction___.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C6 · MECHANICAL SYSTEMS

# Machines at work

★★☆
Grade 5

**Learning goals**

- Explain how simple machines help.
- Analyse a mechanical system.

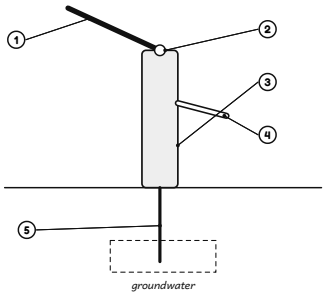
**1** **Rahim pushes a heavy box up a ramp instead of lifting it straight up. What does he gain, and what does he give up?** [2]

He needs a smaller force, but he must push the box a longer distance.

**2** **Why is it easier to open a door by pushing near the handle than near the hinges?** [2]

The door is a lever with the hinge as the pivot; pushing further from the pivot needs less force.

**3** **Analyse the hand tube-well.** [4]




**a)** Which simple machine is the handle?

a lever (the pivot is near the pump body)

**b)** Why is the handle long?

A long handle means the effort is far from the pivot, so less force is needed to lift the water.

**c)** What force starts the pump, and what is the output?

a person pushing down on the handle; water flows out of the spout

**4** **Name three simple machines on a bicycle or rickshaw and what each does.** [6]

| Part | Simple machine | What it does |
|---|---|---|
| pedal and crank | lever / wheel and axle | turns the push of the feet into turning |
| wheels | wheel and axle | help it roll with less friction |
| brake handles | lever | make a big squeezing force on the wheel |

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C6 · MECHANICAL SYSTEMS

# How does distance from the pivot change the effort?

★★☆
Grade 5

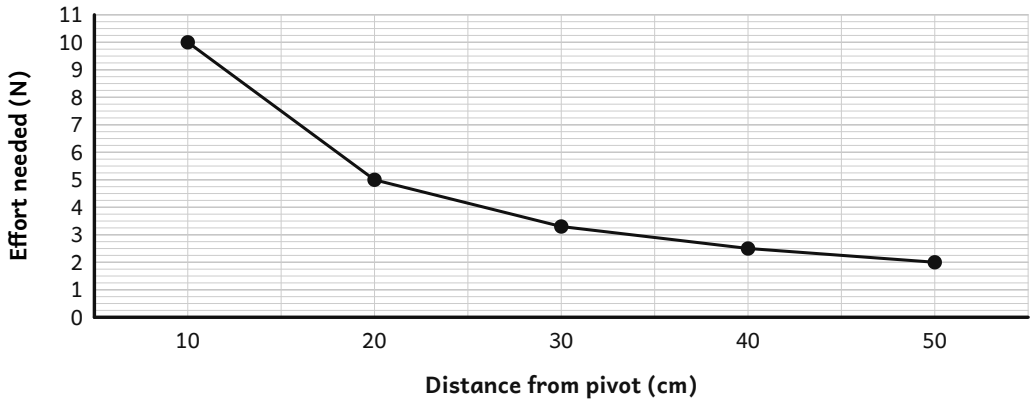
**Learning goals**

- Measure force with a forcemeter.
- Find a pattern.
- Make a prediction.

Class 5A put a 10 N load on a ruler lever, 10 cm from the pivot. They used a **forcemeter** to measure the effort needed to lift it, pulling at different distances from the pivot on the other side. (N = newton, the unit of force.)

**1** **Here are the results. Plot them as a line graph.** [4]

| Distance of effort from pivot (cm) | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|
| Effort needed (N) | 10 | 5 | 3.3 | 2.5 | 2 |




**2** **Describe the pattern.** [2]

The further the effort is from the pivot, the less force is needed.

**3** **Predict the effort needed at 60 cm.** [1]

about 1.7 N (a little less than 2 N)

**4** **Name two things the class kept the same.** [2]

the load (10 N), the load's distance from the pivot, the same ruler and pivot

**5** **Use the results to explain why a long crowbar helps to lift a heavy stone.** [1]

A long crowbar lets you push far from the pivot, so you need much less force to lift the heavy stone.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C6 · MECHANICAL SYSTEMS

# The rickshaw: a city on three wheels

★★★
Grade 5

**Learning goals**

- Analyse a real mechanical system.
- Explain how forces power and stop it.
- Consider people, jobs and technology change.

Science · Design · Economics · Ethics · Art

Dhaka has hundreds of thousands of **cycle rickshaws**, famous for their colourful hand-painted art. A rickshaw is a **mechanical system**: the puller pushes the **pedals** (levers), which turn a **crank** and **gear wheel**. A **chain** carries the turning to a small gear on the back axle, turning the big **wheels**. **Brake levers** press pads against the wheels, and **friction** slows them down.

Pulling a rickshaw is hard work, especially with three passengers or on a bridge. In recent years many rickshaws have been fitted with **batteries and electric motors**. They are faster and easier for the puller, but they can be dangerous at high speed, the batteries must be charged with electricity, and old batteries can pollute if not recycled.

Based on reports about rickshaws and battery rickshaws in Bangladesh.

**1** **Complete the table to show how the rickshaw works as a system.** [6]

| Part | Simple machine | Job in the system |
|---|---|---|
| pedals and crank | lever / wheel and axle | turn the puller's push into turning |
| chain and gears | wheel and axle (gears) | carry the turning to the back wheels |
| wheels and axles | wheel and axle | let the rickshaw roll |
| brake levers | lever | press the brakes hard on the wheel to stop it with friction |

**2** **Which force starts the rickshaw moving and which force stops it?** [2]

Starts: the puller's push on the pedals (muscle force). Stops: friction from the brakes (and air resistance).

**3** **What might each person think about battery rickshaws?** [3]

| Person | View |
|---|---|
| a rickshaw puller | Less tiring and I can earn more, but the battery costs money. |
| a traffic police officer | They go too fast in traffic and can cause accidents. |
| an environmental scientist | No smoke, but the batteries need electricity and safe recycling. |

**4** **Design. Suggest one improvement to make the cycle rickshaw easier to pull or safer. Explain how it uses forces or simple machines.** [2]

e.g. lower gears for bridges (smaller force needed); better brakes with longer levers; lighter frame; reflectors and lights for safety.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C6 · MECHANICAL SYSTEMS

# Show what you know

★★★
Grade 5

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| S | A | X | L | E | V | E | S | B | V |
|---|---|---|---|---|---|---|---|---|---|
| E | H | R | L | P | I | V | O | T | P |
| A | Y | W | A | L | N | E | L | S | U |
| G | D | H | F | M | E | E | H | P | L |
| G | S | E | R | O | P | V | O | V | L |
| I | C | E | Y | L | R | O | E | V | E |
| H | R | L | B | S | C | C | H | R | Y |
| K | E | N | K | N | H | E | E | O | K |
| N | W | L | V | K | W | E | D | G | E |
| I | M | K | N | M | F | C | E | S | U |

**axle force lever pivot pulley ramp screw wedge wheel**

**2** **Which simple machine is a knife?** [1]

- ○ pulley
- ● **wedge**
- ○ screw
- ○ wheel and axle

**3** **What does a single fixed pulley do?** [1]

- ○ makes the load lighter
- ● **changes the direction of the force**
- ○ stops friction
- ○ makes the rope longer

**4** **Design a machine to lift a bucket of sand to the top of a wall. Draw and label which simple machines it uses.** [3]

My machine

e.g. a pulley at the top with a rope; or a lever like a shadoof; labels showing effort, load and pivot or pulley.

**5** **Challenge. A see-saw has a heavy child on one side and a light child on the other. How can they balance? Explain using the pivot.** [2]

The heavy child sits closer to the pivot and the light child sits further away; the light child's smaller force acts further from the pivot.